WATERWATCH · PRODUCT REQUIREMENTS AND SPECIFICATIONS

WaterWatch product requirements and specifications

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: PR-001 restated as ±30 % above 1.0 mg/L (O5), verified against the shared DPD photometer (O2), not met to met on paper; PR-008 counts the hourly schedule only (O4); PR-010 with the operator's photometer (O2)
0.7	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: becomes WWT-PRS-001 (replaces WWT-REQ-001); R1 to R13 carried to PR-001 to PR-013 with the same targets, verification and status, old numbers in Was; tolerance, units and the user requirement each traces to added; PR-014 to PR-016 added for user needs that had no requirement; R9 battery figures follow the one-cell fix (WWT-DEC-001 F1); status from WWT-CAL-001 v0.6 after the independent review (PR-001 not met at 1.5 mg/L, O5)
0.6	2026-10-02	Amish Chadha	

			R7 and R12 figures from WWT-CAL-001 v0.4 (status light); no status changed
0.5	2026-10-02	Amish Chadha	Status from WWT-CAL-001 v0.3 for the constructable design (WWT-DDR-003); R12 stated against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	For TRL 3, status of every requirement from WWT-CAL-001; decisions from WWT-DDR-001 recorded; targets unchanged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWatch product requirements and specifications

What the design must do, each requirement traced to a user need in WWT-URS-001. This document replaces the requirements WWT-REQ-001 (v0.6): every R number is carried to a PR number with the same target, verification and TRL 3 status, and the old number is in the "Was" column. The targets are proposals for review with a co-design partner, not yet validated needs. Status comes from the calculation note WWT-CAL-001 v0.7. On paper, of the 16 requirements, 3 are at risk (PR-002 turbidity, PR-009 outdoor survival and PR-010 caretaker maintenance), and PR-012 cost is over the value-engineering target by USD 25. PR-001 free chlorine was not met at 1.5 mg/L after the independent review of the STEM check corrected the error budget; Amish decided on 2026-10-25 ("i concur with your recommendations") to restate it as ±30 % above 1.0 mg/L (WWT-DEC-001 O5) and to use a shared DPD photometer as the monthly reference (O2), so it is now met on paper (WWT-CAL-001 E11, E12). On paper 12 of the 16 are met (6 by calculation and 6 by design). Method: STANDARDS section 22. The traceability matrix (WWT-RTM-001) is generated by `python .kit/trace.py`.

Table 1. Product requirements and specifications (not met and at-risk items first).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-010	UR-009	Usability	Caretaker maintenance	The unit shall stay within PR-001 with one visit a month or less by a caretaker or operator with the operator's shared DPD photometer (WWT-DEC-001 O2) and no special tools: clean the cell, compare, adjust calibration from a phone	30	max per visit; interval 1 month min	min	Task analysis (WWT-CAL-001 J3), TRL 3; timed visits, TRL 5	At risk
PR-002	UR-002	Performance	Turbidity	The unit shall measure turbidity from 0 to 100 NTU and report "above 100 NTU" beyond	±0.5 or ±10 %, whichever is greater	After the 30 s settling wait	NTU	Bubble and settling calculation (WWT-CAL-001 F), TRL 3; stabilised standards, TRL 4	At risk
PR-009	UR-006	Physical architecture	Outdoor survival	Electronics IP65 or better; ambient 0 to 45 °C in direct sun; UV-stable parts; flow cell opaque to daylight; battery charging blocked below 0 °C and above 45 °C	45	Battery charging for at least 4 sun hours on a clear 45 °C day	°C	Thermal estimate (WWT-CAL-001 C), TRL 3; logged enclosure temperature, TRL 4	At risk

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-012	UR-011	Packaging and transport	Parts cost	Parts cost per base unit, excluding the shared calibration kit, airtime and the pH probe variant	300	Value-engineering target, not a limit	USD	Priced BOM (WWT-CAL-001 K1), TRL 3	Over target
PR-005	UR-003	Software	Alerts	SMS to up to three numbers within 15 min of a confirmed crossing (two readings, the second 15 min after the first): free chlorine below 0.2 mg/L, turbidity above 5 NTU, device fault or low battery; thresholds adjustable per site	15	max, where there is coverage	min	Latency calculation (WWT-CAL-001 H), TRL 3; network test, TRL 4	Met on paper
PR-006	UR-004	Software	Data	Upload readings at least every 4 h over LTE-M, NB-IoT or 2G (LoRaWAN variant); keep 90 days on board; open CSV or JSON; no cloud service required	1	max per month of uploads	MB	Data budget (WWT-CAL-001 G), TRL 3	Met on paper
PR-007	UR-005	Electrical	Solar autonomy	Run 7 days or more without sun	7		days	Energy budget (WWT-CAL-001 A, B),	Met on paper

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				from full, and refill in 3 clear days or fewer		min; refill 3 clear days max		TRL 3; logged energy, TRL 4	
PR-001	UR-001	Performance	Free chlorine	The unit shall measure free chlorine at the tap from 0 to 2.0 mg/L at pH 6.5 to 8.5 and 5 to 40 °C; sites whose pH is above 7.5 or moves more than 0.2 between visits use the pH probe variant (WWT-DDR-002)	±0.2 mg/L or ±25 %, whichever is greater, up to 1.0 mg/L; ±30 % above 1.0 mg/L (restated by Amish, 2026-10-25, WWT-DEC-001 O5)	Before sensor drift; at least 0.15 mg/L left for drift at 25 °C	mg/L	Analysis (WWT-CAL-001 E), TRL 3; side by side with the shared DPD photometer (O2), TRL 4	Met on paper: 8 of 8 cases with the photometer as the monthly reference, worst ±0.411 against ±0.450 mg/L (WWT-CAL-001 E12)
PR-008	UR-008	Performance	Water use	Water flushed to drain at hourly sampling; the hourly schedule only is counted, not the rare alert-driven confirming readings (decided by Amish, 2026-10-25, WWT-DEC-001 O4)	20	max, at the regulator's upper tolerance	L/day	Flow calculation (WWT-CAL-001 D), TRL 3; measured, TRL 4	Met on paper
PR-011	UR-010, UR-007	Usability	Installation	Two people with hand tools install the unit in 2 h or less on installation day, with the	120	max on installation day	min	Task analysis (WWT-CAL-001 J1, J2), TRL 3; timed install, TRL 5	Met on paper

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				footing cast on the survey visit; one tee and isolation valve on the riser					
PR-003	UR-013	Performance	Water temperature	The unit shall measure water temperature from 0 to 50 °C	±0.5	Sensor datasheet	°C	Inspection of the datasheet, TRL 3	Met by design
PR-004	UR-016	Software	Schedule	One reading an hour by default, configurable from 15 min to 24 h	15 to 1440	Energy at 15 min checked (WWT-CAL-001 A5)	min	Analysis, TRL 3; firmware demonstration, TRL 4	Met by design
PR-013	UR-012	Software	Measurements, not verdicts	Messages state what was measured, when and the threshold crossed; the device never says the water is "safe"	0	No message contains a safety verdict	messages	Inspection of the alert templates (WWT-DEC-001), TRL 3	Met by design
PR-014	UR-007	Safety	Supply protection	The sample line shall have an isolation valve and two check valves in series, the cell outlet an open air-break vent above the outlet, and wetted parts shall be	60	Vent height above the outlet, min	mm	Inspection of the drawing WWT-DWG-001 and BOM, TRL 3; backflow test, TRL 4	Met by design

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				food-contact grades					
PR-015	UR-014	Usability	Open files	The CAD source, STEP, drawings, BOM, calculations and build plan shall be published under CERN-OHL-S-2.0 and the software under MIT	100	Share of design files published	%	Inspection of the repository and REUSE lint, TRL 3	Met by design
PR-016	UR-015	Safety	Battery and chemical handling	The battery shall have a protection board, a fuse at the cell and charging only from 0 to 45 °C; calibration shall use prepared, stabilised standards	0 to 45	Charge window	°C	Inspection of the BOM and build plan, TRL 3; charger test, TRL 4	Met by design

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Notes on status:

- **PR-001.** The budget of WWT-CAL-001 v0.5 met the target in all 8 cases at 25 °C before drift (E6). With the stale-water bias added linearly, as the independent review asked, 6 of 8 cases are met; at 1.5 mg/L it misses by 0.004 mg/L at pH 7.5 and 0.017 mg/L at pH 8.5 (E9), and when the sensor is calibrated at 0.5 mg/L and reads 1.5 mg/L only 1 of 4 high cases is met (E10). At 0.2 to 0.5 mg/L, where the alerts are set, it was met with at least 0.13 mg/L left for drift. Amish restated the target as ±30 % above 1.0 mg/L on 2026-10-25 (WWT-DEC-001 O5), which the comparator budget meets in all 8 cases (E11); with the shared DPD photometer (±0.03 mg/L ±3 %, WWT-MKT-001) as the monthly reference (O2) all 8 are met, calibrated at the reading level or at 0.5 mg/L, with at least 0.15 mg/L left for drift (E12). Status not met to met on paper. The drift of the low-cost sensor over a month is unknown (WWT-STM-001 S12, S14).
- **PR-005.** 13.0 min from a confirmed crossing to the SMS with three tries; from the event itself up to 92 min with all tries (WWT-CAL-001 H4).
- **PR-006.** 0.36 MB a month with TLS session resumption (WWT-DEC-001 F7); a full handshake every session would use 1.22 MB (G3).

- **PR-008.** 19.8 L/day at the regulator's tolerance on the hourly schedule; at a 0.5 bar site with the 110 s flush (F5) 15.4 L/day; confirming readings add about 0.25 L/day; Amish decided on 2026-10-25 that PR-008 counts the hourly schedule only, as the requirement now says (WWT-DEC-001 O4).
- **PR-014.** Two check valves in series (supply train and cell inlet, WWT-DEC-001 F8) and the normally closed valve stand between the cell and the supply; a verifiable backflow device or an inlet air gap is to assess under the local code (WWT-REG-001 R4).
- **PR-009.** With the sun shield the inside peaks at 50.8 °C on a 45 °C day, and charging is allowed for 5.4 sun hours (WWT-CAL-001 C4); the shield factor is assumed. The one-cell battery (WWT-DEC-001 F1) lowers the thermal mass from 1,460 to 1,305 J/K and leaves the peaks unchanged to 0.1 °C.
- **PR-012.** USD 325 per base unit, USD 25 over the target (WWT-CAL-001 K1); USD 385 to 425 at a pH probe variant site (K2); the second check valve (WWT-DEC-001 F8) added USD 3.

Assumptions

- Site: public tapstand on a piped, chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi) at the riser. Tapstands come first; tank outlets and kiosks later (WWT-DDR-001, D8).
- Alert thresholds follow the field guidance summarized in WWT-PRB-001: at least 0.2 mg/L free chlorine at the point of delivery, and chlorination losing effectiveness as turbidity rises. The defaults of 0.2 mg/L and 5 NTU were decided by Amish on 2026-09-25 (WWT-DDR-001, D6) and remain adjustable per site.
- The caretaker's DPD test is the field reference for free chlorine. The v0.3 target of ±0.1 mg/L matched the resolution of common DPD comparator kits, which WWT-CAL-001 showed made it as tight as the reference itself (the comparator alone takes ±0.07 mg/L); R1 was relaxed to ±0.2 mg/L or ±25 % for that reason (WWT-DDR-002).
- pH affects the split between hypochlorous acid and hypochlorite, and amperometric sensors respond mainly to hypochlorous acid. R1 is assessed with a site pH entered at installation and checked monthly (WWT-DDR-001, D2), assumed to move no more than ±0.2 between visits. Under WWT-DDR-002 the pH probe variant (read to ±0.1) is fitted at sites whose pH is above 7.5 or moves more than 0.2 between visits; the cell lid carries a plugged spare port for it.
- Budget: the $300 in `project.yaml` is a value-engineering target for one base unit's parts, not a spending limit (Amish, 2026-10-01). A shared calibration kit (about $60), airtime (about $1 to $3 per month) and, at variant sites only, the pH probe and interface (about $60 to $100) are listed separately; all are estimates.
- Installation: the pole footing is dug and cast on the survey visit, so the pole is set and cured before installation day (WWT-DDR-002).

> **Safety:** Meeting these requirements does not make the water safe to drink or the unit safe to install. The sample line connects to a pressurized drinking water supply and the unit contains a lithium iron phosphate battery; see WWT-PRC-001, Safety.